



Analyzing OTT Platform Viewing Patterns to Support Data-Driven Strategy

A comprehensive analysis of user engagement, retention, and content performance across NetStream, AmazePrime, and DizPlay+

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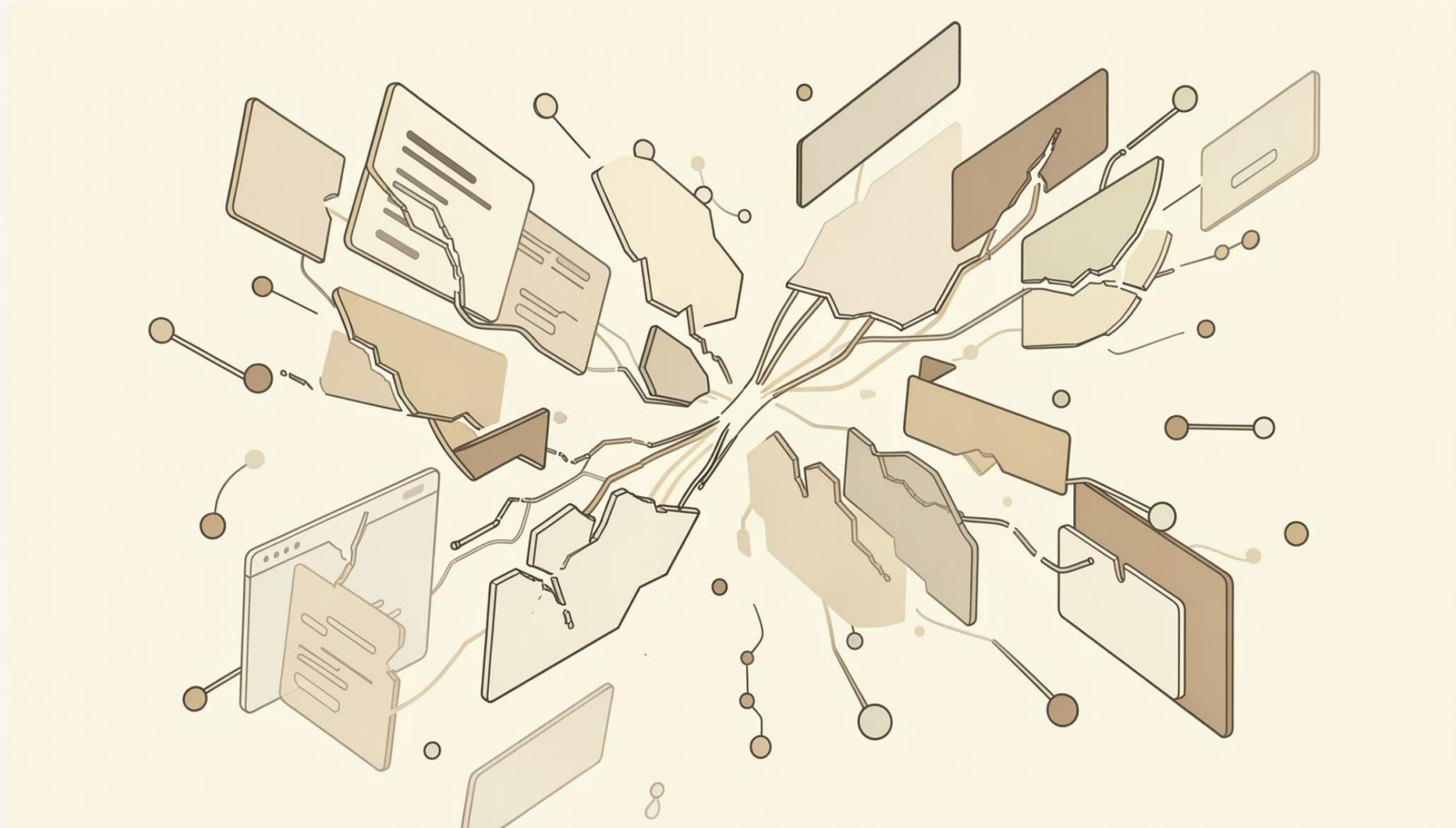
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Introduction

- Evaluates NetStream, AmazePrime, and DizPlay+ performance comprehensively.
- Integrates metadata, profiles, activity, subscriptions, feedback.
- Examines viewing behavior across diverse audience segments.
- Supports retention, engagement, and growth optimization strategies.
- Delivers a unified, enterprise-wide streaming performance perspective.
- Enables rigorous platform benchmarking and strategic comparison.
- Transforms complex data into actionable business intelligence.

Problem Statement



The company currently **lacks a unified understanding** of user engagement and retention across its platforms. Fragmented data across content, users, subscriptions, and viewing activity makes it difficult to answer critical business questions:

Without this clarity, data-driven decisions for improving user experience and expanding market presence remain out of reach.

→ What drives user retention?

→ Which content performs best — and where?

→ How does user behavior vary across platforms?

Project Objectives

Seven interconnected goals guide this analysis — from surface-level behavior to predictive modeling and strategic action.



Content Discovery

Analyze how users find content via genres and recommendation sources



Viewing Habits

Understand what, when, and how users watch across devices and time



Subscription Insights

Track plan performance, cancellations, and user lifetime value



Ratings & Feedback

Explore content ratings and audience sentiment across platforms



Statistical Analysis

Apply t-test, z-test, and chi-square to validate data relationships



ML Modeling

Build regression, classification, and clustering models for predictions



Strategic Recommendations

Translate insights into targeted strategies for maximizing user lifetime value and competitive positioning

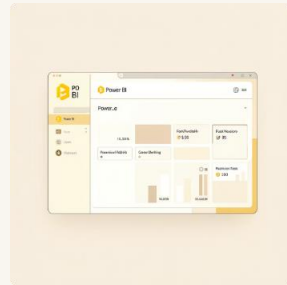
Tools & Technologies

A modern analytics stack — combining spreadsheet analysis, visual dashboarding, programmatic modeling, and structured data management.



Excel

Organize, clean, and perform foundational analysis using formulas and charts



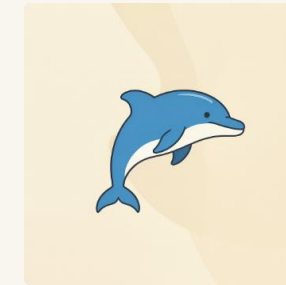
Power BI

Build interactive dashboards and compelling visual narratives for stakeholders



Python

Execute statistical analysis, visualization, and machine learning pipelines



MySQL

Store, manage, and retrieve structured data efficiently with SQL queries



Customer Churn & Engagement Dashboard



TOTAL USERS
38,481

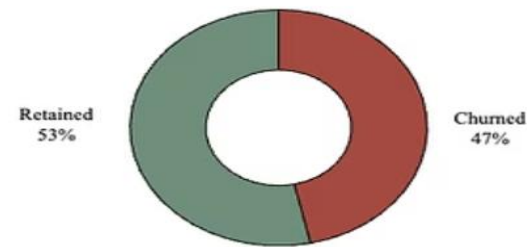
CHURN RATE
46.5%

AVG WATCH TIME
180 min

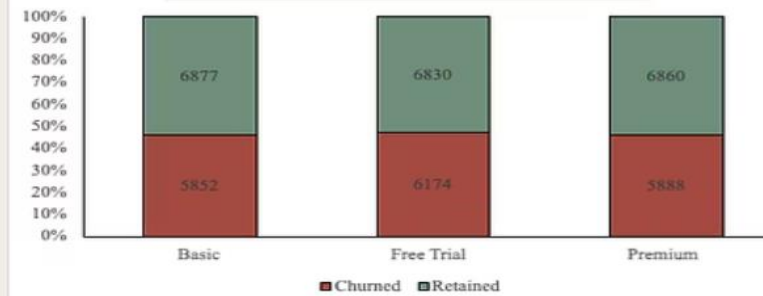
AVG COMPLETION RATE
52.2%

REWATCH RATE
49.8%

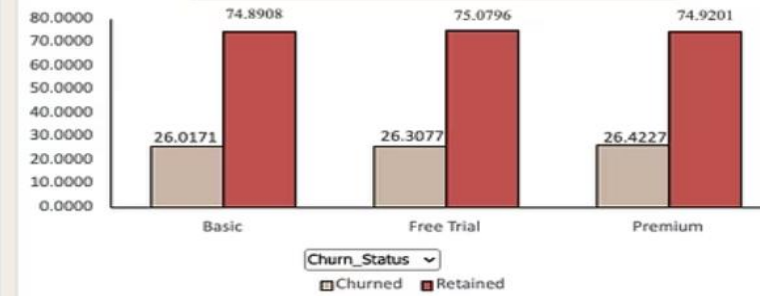
Churn vs Retention Distribution



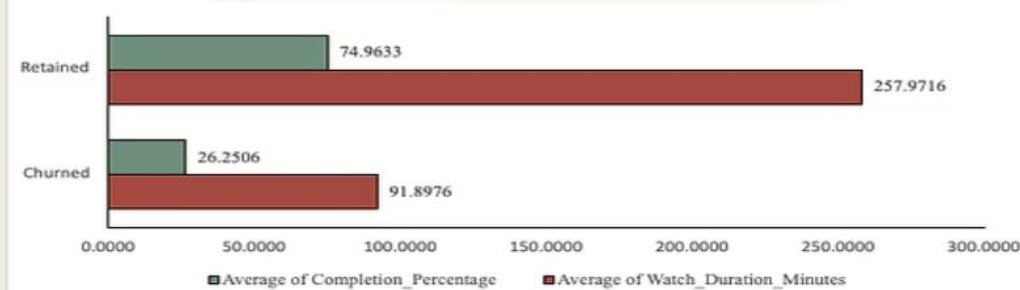
Retention Rate by Subscription Type



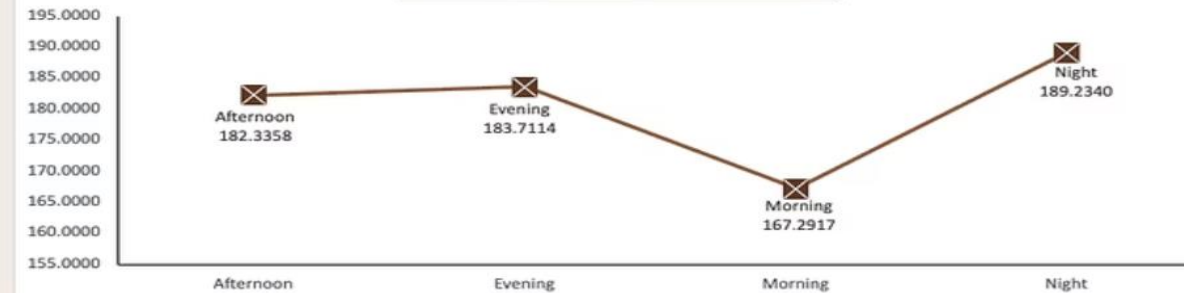
Engagement Metrics by Subscription



Engagement Comparison: Churned vs Retained



User Engagement by Time of Day



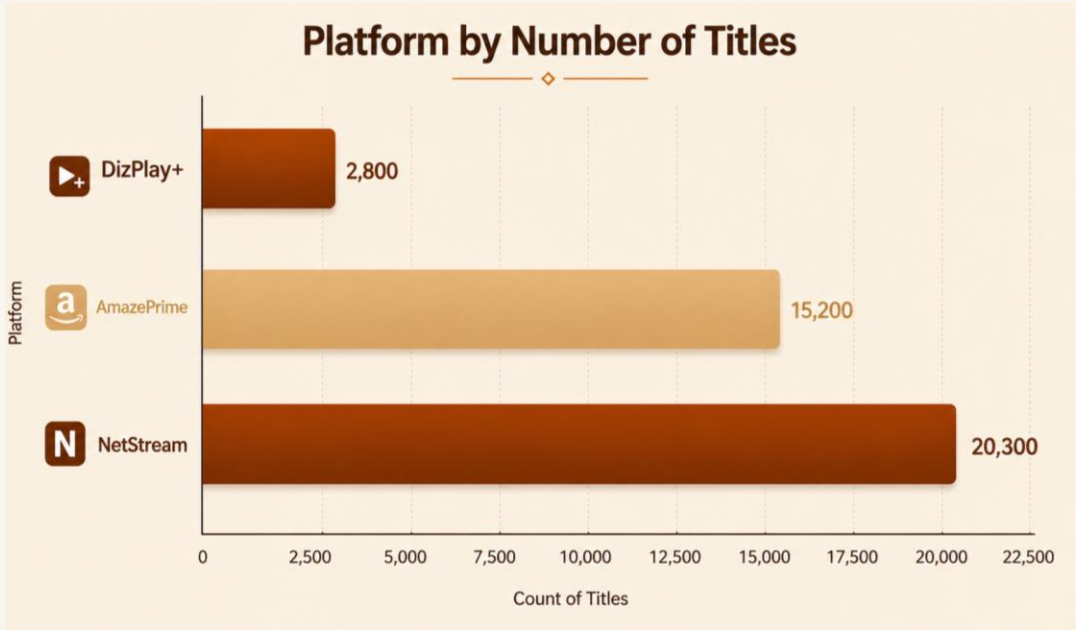
Key Insights

- Low-tier plans show higher churn
- Retained users have significantly higher watch time
- Evening hours drive peak engagement

Recommendations

- Improve content for basic plan users
- Target high-engagement time slots
- Focus retention strategies on low-usage segments

High Engagement Content Insights



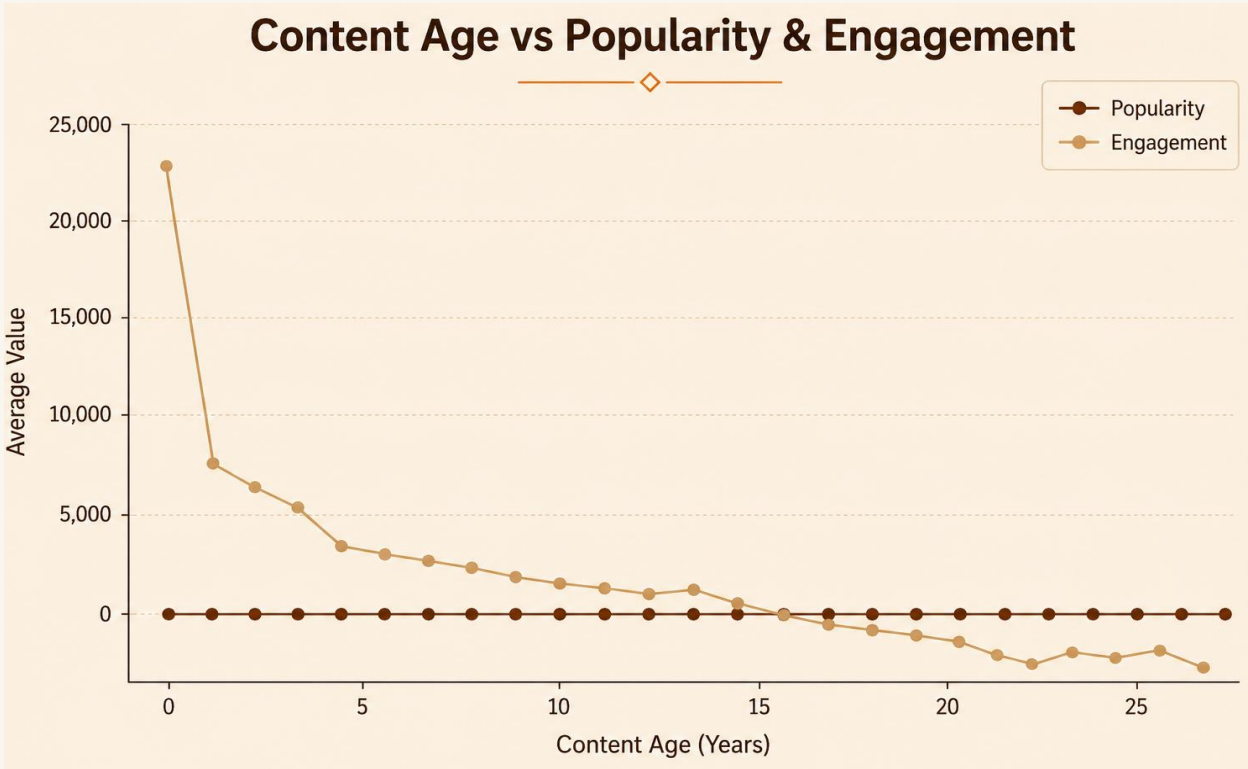
Platform by Title Count

This chart provides a clear comparison of content libraries across platforms, helping identify market leaders in title availability. It also highlights content gaps, supporting strategic decisions in competitive analysis and platform positioning.

Content Age vs. Engagement

Popularity and engagement **drop sharply** after year 1. Titles older than 5 years consistently underperform.

^ Old content is a low performer — fresh releases drive the buzz.



NetStream

Largest library — prime target for deep engagement analysis

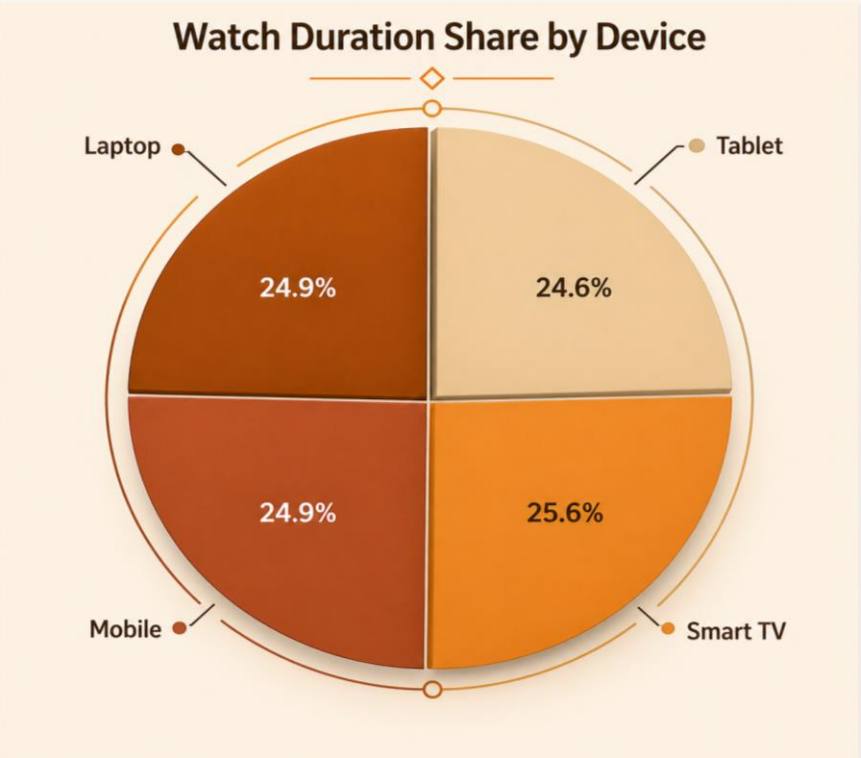
AmazePrime

Strong mid-tier presence — room to grow with fresh content

DizPlay+

Smaller library — content investment needed to compete

Device & Content Length: What Drives Engagement



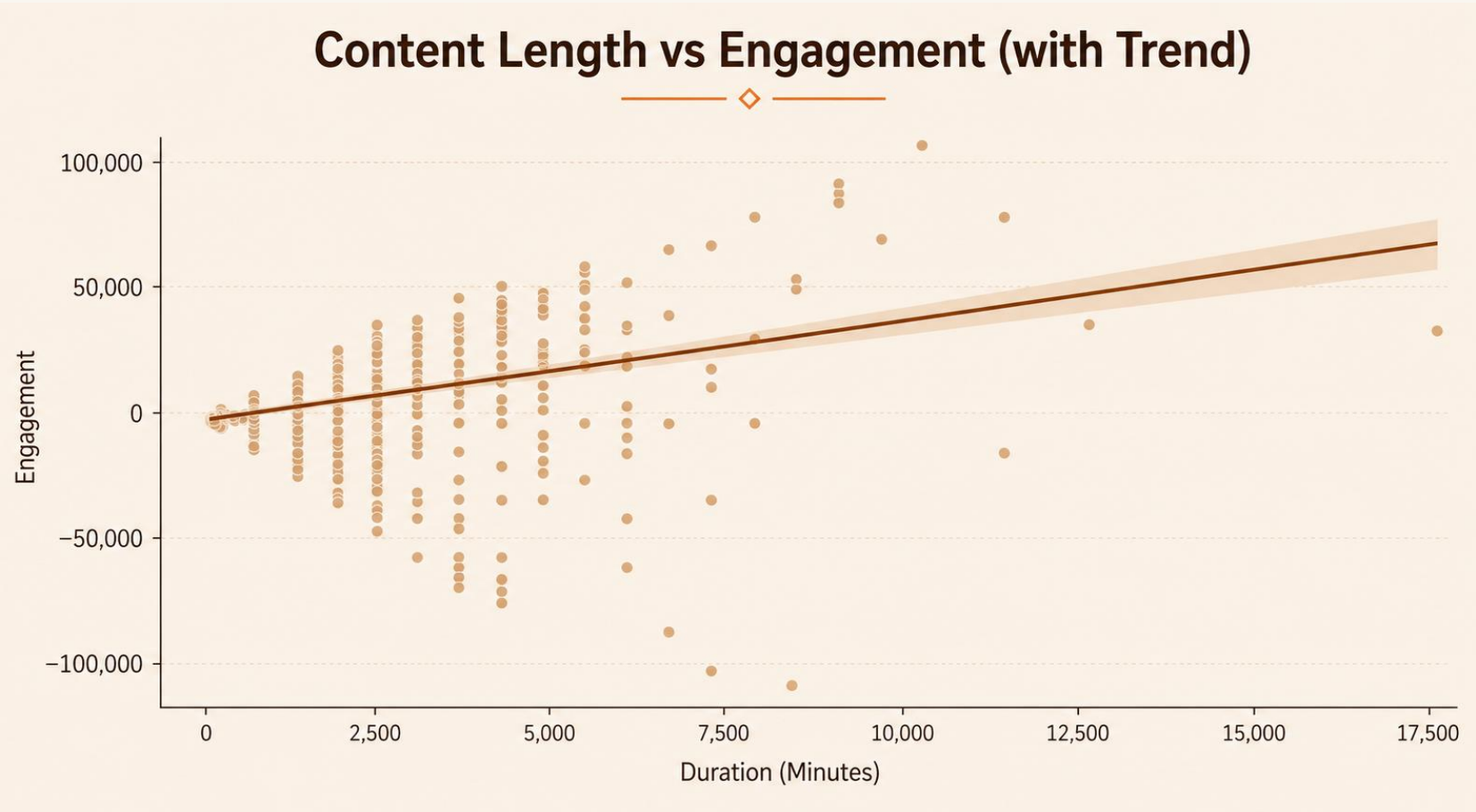
Content Length vs. Engagement

A clear upward trend: **longer content drives higher engagement scores**. Investing in long-form programming pays off.

- ☑ Users who start longer content are more likely to interact, rewatch, and retain.

Watch Duration by Device

Remarkably even split across all four devices — each claims roughly **25%** of total watch time.



Content Discovery

An overview of what users are watching — top genres, platform distribution, total content, and how completion rates compare to rewatch behavior across the catalog.

Expand Underserved Genres

Romantic TV Shows generated the highest watch time (3.8K mins)

Address Completion Drop-Off

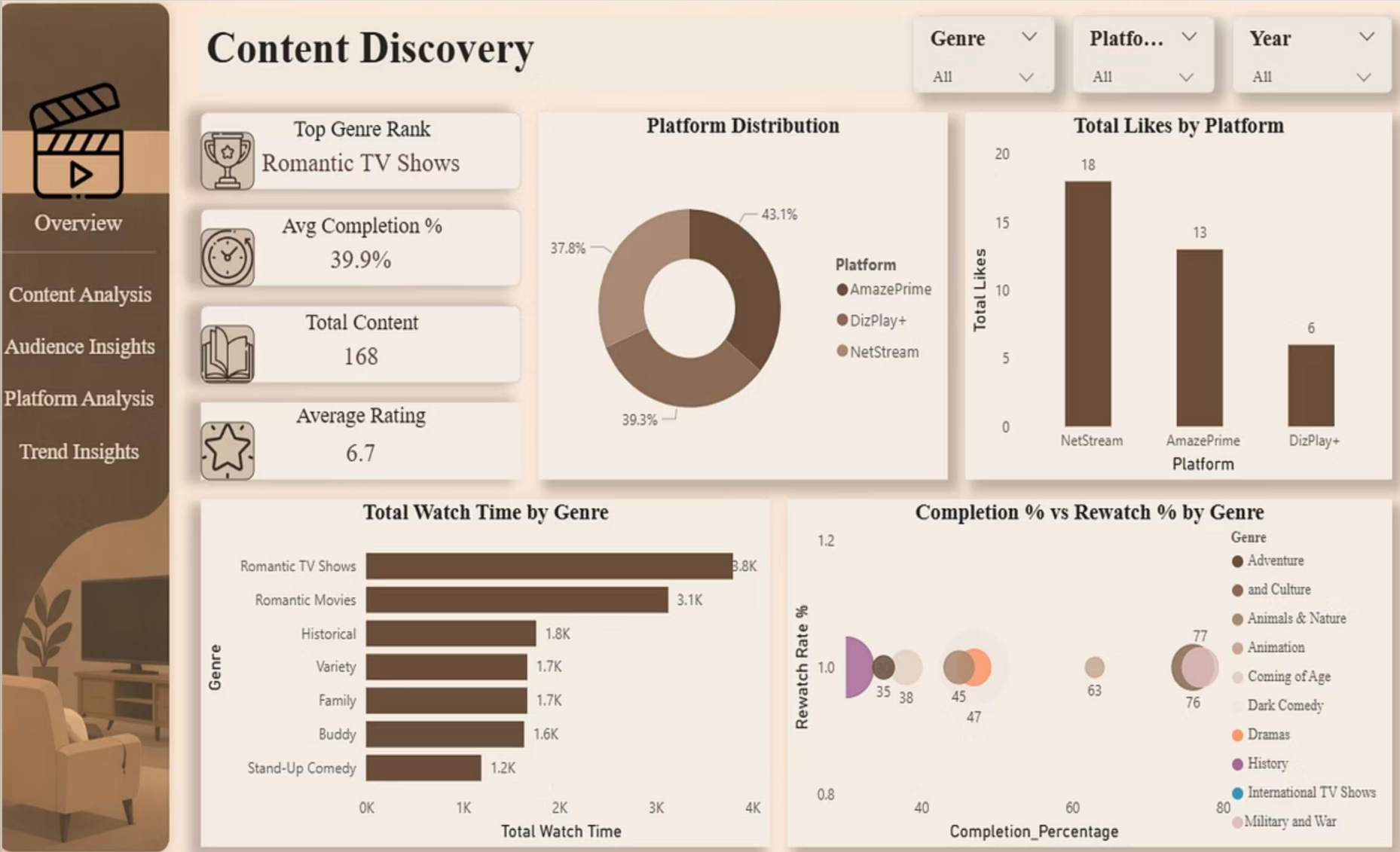
Average completion rate is 39.9%

Strengthen Content Acquisition

NetStream received the highest audience likes (18)
Romantic TV Shows ranked as the top-performing genre

Improve Discoverability

NetStream leads platform share (43.1%)



Viewing Behavior

Understanding *when*, *where*, and *how* users consume content — from peak session times to device preferences and geographic engagement patterns.

Optimize for Peak Afternoon Viewing

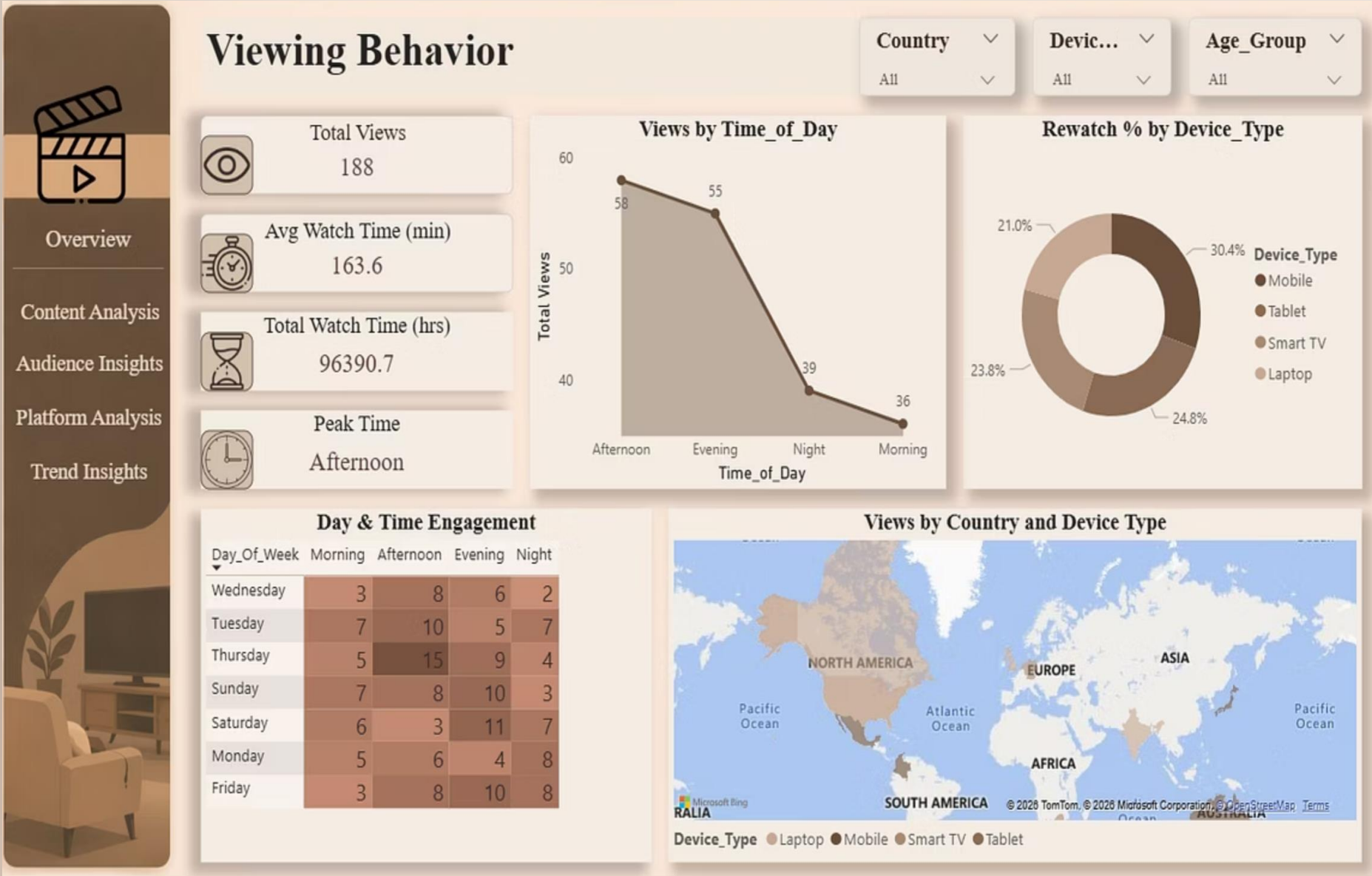
Afternoon recorded the highest engagement (58 views)

Personalize by Time Zone

Mid-week evening sessions show peak activity

Reduce Device Fragmentation

Smart TV users contribute the highest rewatch percentage (25.6%)



Overview

Content Analysis

Audience Insights

Platform Analysis

Trend Insights

Subscription Trends

Total Subscribers
72

Retention Rate
48.6%

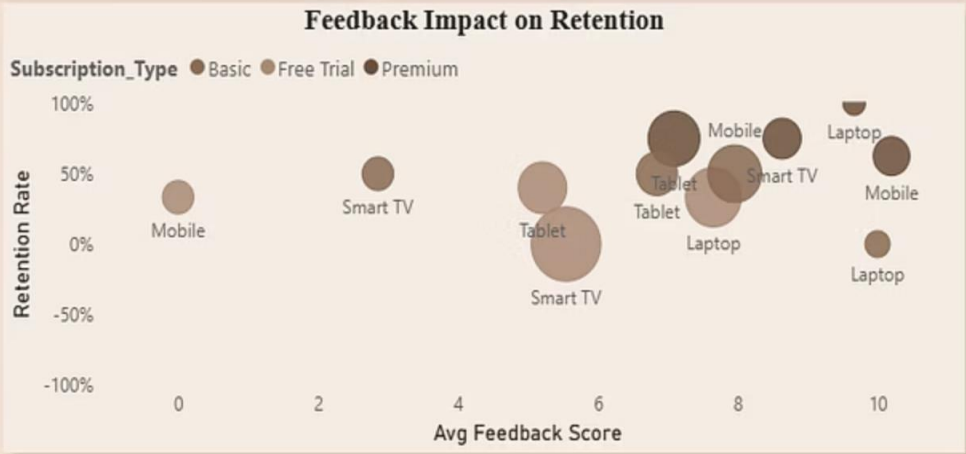
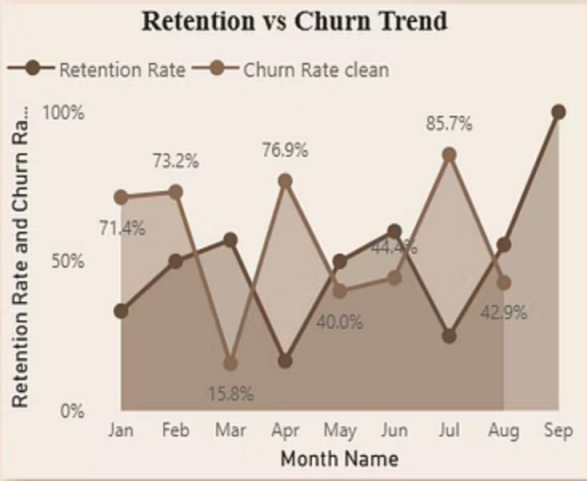
Avg Feedback Score
6.7

Churn Rate
51.6%

Subscription_T...
All

Region
All

Tenure Group
All



Subscription Trends

A look at subscriber health — how retention and churn evolve over time, broken down by tenure group and subscription type, and how feedback scores influence renewal.

Focus on High-Value Subscribers

Retention rate currently stands at 48.6%

Average subscriber feedback score: 6.7

Monitor Long-Term Churn

Highest churn observed in the 12+ Months tenure group

Accelerate Trial Conversion

Free Trial users contribute a smaller subscriber share

Use Feedback to Drive Retention

Higher feedback scores align with stronger retention



Ratings & Feedback

Rating ▼
All ▼

Platform ▼
All ▼

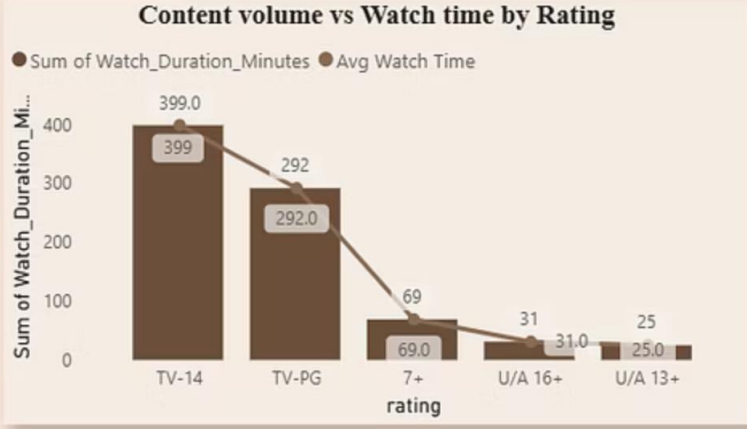
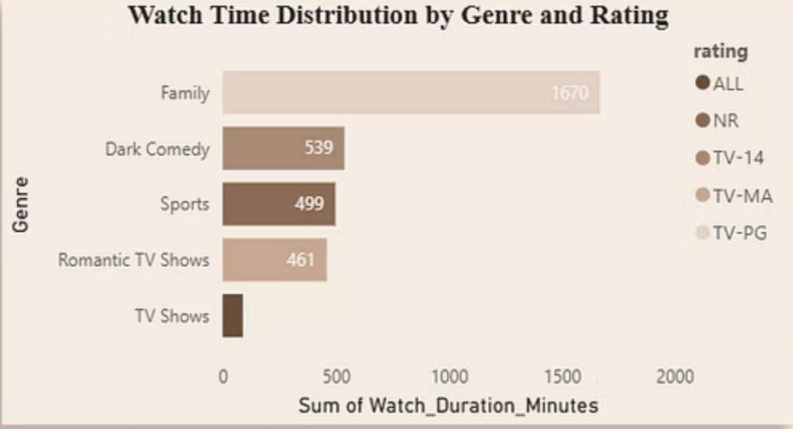
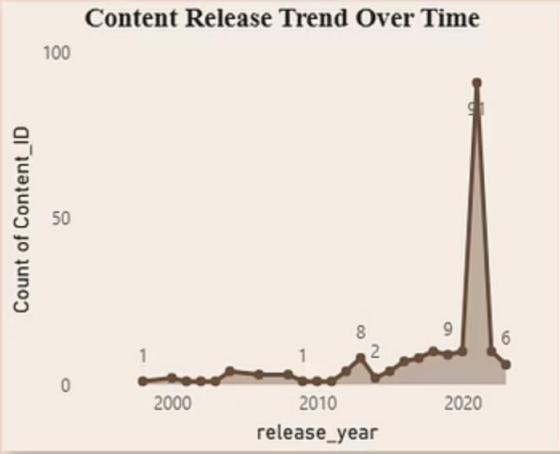
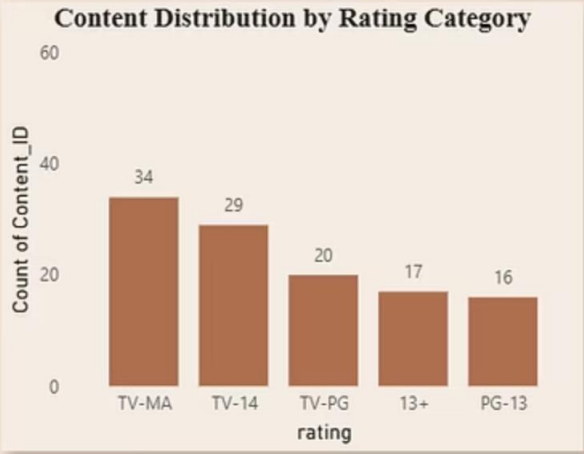
Type ▼
All ▼

Positive Rating %
49.5%

Most common Rating
13+

Mature Content %
1.6%

Content Diversity
19



Ratings & Feedback

Audience sentiment and content rating distribution — examining how rating categories, release timelines, and watch time intersect to reveal what resonates most.

Optimize Audience Segmentation

TV-MA is the most common content category (34 titles)

Fill Content Gaps

Content diversity score stands at 19
Family genre generated the highest watch duration

Scale What Works

TV-14 titles achieved the highest watch duration (399 mins)

Use Combined Signals

Positive audience rating percentage: 49.5%

Statistical Hypothesis Validation

Three statistical tests were applied to the OTT dataset to validate behavioral assumptions — examining gender differences, watch duration benchmarks and churn associations.

1

T-Test: Gender vs. Watch Duration (Alpha Value : 0.05)

Result: Fail to reject H_0 · T-stat: -0.036
· P-value: 0.971 (30 Samples)

No statistically significant difference in average watch duration between male and female users. Gender does not predict session length.

2

Z-Test: Gender vs. Watch Duration (Alpha Value : 0.05)

Result: Fail to reject H_0 · Z-stat: -0.61
· P-value: 0.54 (62 Samples)

There is no statistically significant difference between the average watch durations of the two groups being compared.

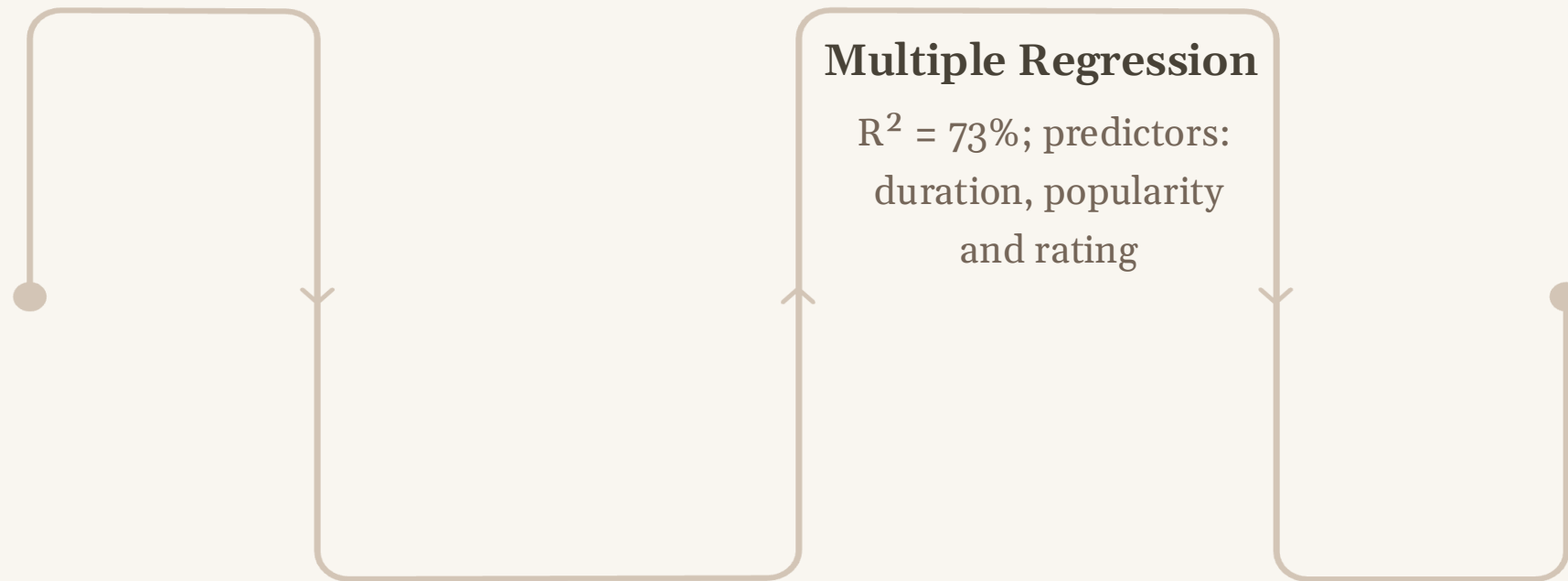
3

Chi-Square: Gender vs. Churn Behavior (Alpha Value : 0.05)

Result: Reject H_0 · $\chi^2 = 55.82$ · P-value: 0
A significant association exists between gender and churn flag — meaning gender is a meaningful predictor of whether a user churns or renews.

Machine Learning: Regression Model

Multiple Linear Regression is a statistical technique used to predict one dependent variable using two or more independent variables.



Multiple Linear Regression

Can user engagement be predicted using duration, popularity and rating with multiple linear regression, and how well does the model perform?

$R^2 = 73\%$ shows the model predicts user engagement well. Popularity drives engagement the most, while duration has a small effect and rating/year have negative impact.

Machine Learning: Classification & Clustering

Beyond regression — three models tackle user renewal prediction and behavioral segmentation, with KNN and K-Means outperforming the baseline logistic approach.



Logistic Regression

Can user renewal status be predicted based on demographic and subscription features using logistic regression, and how well does the model perform?

Accuracy: **69%**



K-Nearest Neighbor (KNN)

Can user renewal status be predicted using a K-Nearest Neighbors (KNN) classification model based on multiple features in the OTT dataset, and how well does the model perform?

Accuracy: **68%**



K-Means Clustering

Can OTT users be segmented into distinct groups based on watch duration and engagement levels using K-Means clustering, and how well are the clusters formed?

Silhouette: **0.581** · DB Score: **0.578**

✔ KNN is the strongest renewal predictor. K-Means clustering unlocks actionable user segments for personalized engagement strategies.



CONCLUSION

- Engagement predicted well ($R^2 = 73\%$) using multiple features
- Watch duration alone explains only 34%
- Popularity is the strongest factor
- Users watch more than 60 mins ($Z = 75$) → high engagement

FINAL CONCLUSION:

The analysis shows that user retention and engagement are strongly influenced by watch time, subscription type, content freshness, and personalized viewing experiences. NetStream appears to be the most preferred platform among users, driven by its larger content library and stronger engagement potential.